

Yi Yuan

Email: yuanyithu@126.com GitHub: github.com/yuanyithu

Male | Ph.D. Candidate in Physics, Tsinghua University

Education

Tsinghua University Ph.D. in Physics

Department of Physics Ranked **1st** in Modern Physics in the Ph.D. Qualifying Exam Principles and Architecture of Quantum Computers (A+) 2023.08 – 2028.06 (Expected) Beijing

Tsinghua University B.S. in Fundamental Sciences, Mathematics and Physics

Department of Physics GPA: 3.89/4.0 (Rank: 18/116) Outstanding Graduate Nominee, Ye Qisun Award 2019.08 – 2023.06 Beijing

Technical Skills

- **Programming Languages:** Python (primary language for daily research and development)
- **AI/ML:** Familiar with Transformer architectures; experienced with JAX and PyTorch; hands-on experience in CNN modeling
- **Data and Computing:** NumPy vectorized computing, multiprocessing, data-driven modeling, and backtesting system development
- **Tools:** Git, Linux servers **Language:** English – proficient in reading and writing scientific literature

Publications

Yi Yuan, Yuanchen Zhao, and Dong E. Liu. “Zeno-Enhanced Probabilistic Error Cancellation with Quantum Error Detection Codes.” *arXiv preprint arXiv:2605.12149*, 2026. [arXiv](https://arxiv.org/abs/2605.12149)

Developed a perturbative QEDC+PEC protocol for near-term noisy quantum devices, using post-selection to reshape the effective noise channel and PEC to cancel residual logical errors.

Research and Projects

Research in Quantum Computing and Quantum Information

Department of Physics, Tsinghua University

2022.09 – Present
Python, JAX, Tensor Network

- **Zeno-enhanced QEDC+PEC protocol** (2024 – 2026, arXiv preprint): Combined quantum error-detecting codes with probabilistic error cancellation for near-term noisy quantum devices; in 200-qubit Iceberg-code logical GHZ preparation, reduced sampling overhead by **3–4 orders of magnitude** relative to standard PEC while maintaining $F \simeq 0.956$
- **Classical Shadow error mitigation** (2024): Used tensor network methods to compute the error-mitigation complexity of classical shadows on noisy quantum circuits with $\log(n)$ depth, and analyzed its scaling with system size
- **Undergraduate thesis: complexity of quantum annealing** (2022.09 – 2023.06): Based on the imaginary-time spacetime instanton action, explained why quantum annealing and classical QMC have matching complexity in ground-state energy search for double-well potentials, and identified the possibility of multi-path quantum speedup

Selected Projects

- **SEVI particle scattering analysis:** Built a PyTorch-based CNN to fit scattering-amplitude parameters, reaching the accuracy of analytical methods without injecting prior knowledge
- **LLM paper summarization tool** (2025): Automatically fetched daily arXiv papers, generated summaries through the Kimi API, and completed local deployment

Honors and Leadership

Honors: 2021 Tsinghua Alumni – Zhang Mingwei Scholarship Multiple Department of Physics scholarships for academic excellence, innovation, and overall achievement in 2020–2021

Leadership: 2023.08–2025.01 Student Counselor, Department of Physics, Tsinghua University 2024–2025 Outstanding Student Leader